ATTRIBUTE INFORMATION

1. **Area (A):** The area of a bean zone and the number of pixels within its boundaries.
2. **Perimeter (P):** Bean circumference is defined as the length of its border.
3. **Major axis length (L):** The distance between the ends of the longest line that can be drawn from a bean.
4. **Minor axis length (l):** The longest line that can be drawn from the bean while standing perpendicular to the main axis.
5. **Aspect ratio (K):** Defines the relationship between L and l.
6. **Eccentricity (Ec):** Eccentricity of the ellipse having the same moments as the region.
7. **Convex area (C):** Number of pixels in the smallest convex polygon that can contain the area of a bean seed.
8. **Equivalent diameter (Ed):** The diameter of a circle having the same area as a bean seed area.
9. **Extent (Ex):** The ratio of the pixels in the bounding box to the bean area.
10. **Solidity (S):** Also known as convexity. The ratio of the pixels in the convex shell to those found in beans.
11. **Roundness (R):** Calculated with the following formula: (4 π A)/(P^2)
12. **Compactness (CO):** Measures the roundness of an object: Ed/L
13. **ShapeFactor1 (SF1):** Relates length to total area.

$$ShapeFactor\ 1 = \frac{Major\ axis\ Length}{Area}$$

14. **ShapeFactor2 (SF2):** Relates width to total area.

$$ShapeFactor\ 2 = \frac{Minor\ axis\ Length}{Area}$$

15. **ShapeFactor3 (SF3):** A variation of circularity based on the major axis.

$$ShapeFactor\ 3 = \frac{4 * Area}{\pi\ *\ Major\ axis\ Length^2}$$

16. **ShapeFactor4 (SF4):** Measures how well the bean fills the ellipse defined by its major and minor axis.

$$ShapeFactor\ 4 = \frac{4 * Area}{\pi * Major\ axis\ Length * Minor\ axis\ Length}$$

17. **Class** (Seker, Barbunya, Bombay, Cali, Dermosan, Horoz and Sira)